Quantum Blackholes, Wall Crossing and Mock Modular Forms

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to

SCHOOL OF PHYSICS
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February 5, 2026

DECLARATION

I, **Tharun Naagesh (Roll No: IMS22252)**, hereby declare that, this report entitled **"Quantum Blackholes, Wall Crossing and Mock Modular Forms"** submitted to Indian Institute of Science Education and Research Thiruvananthapuram towards the partial requirement of **Master of Science** in **Mathematics**, is an original work carried out by me under the supervision of **Dr. Bindusar Sahoo** and has not formed the basis for the award of any degree or diploma, in this or any other institution or university. I have sincerely tried to uphold academic ethics and honesty. Whenever a piece of external information or statement or result is used then, that has been duly acknowledged and cited.

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This is to certify that the work contained in this project report entitled **"My Project Title"** submitted by [**Full Name**] (**Roll No: [Roll Number**]) to Indian Institute of Science Education and Research, Thiruvananthapuram towards the partial requirement of [**Master of Science/ Doctor of Philosophy**] in [**Department Name**] has been carried out by [him/her/them] under my supervision and that it has not been submitted elsewhere for the award of any degree.

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Write about the people and the things you are indebted to in fullfilling this project

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ABSTRACT

If you have to structure it as objective, methods, results, and conclusions, do it that way.

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Chapter 1

Prelimnaries

1.1 Primer on Fourier Series

1.1.1 Discrete Fourier Transform

Let

$$\omega := e^{2\pi i/n} \tag{1.1.1}$$

be the nth root of unity. We then have the following identity:

$$\sum_{k=0}^{n-1} w^k(b-a) = n\delta(a,b) \tag{1.1.2}$$

This identity can be written as:

$$M = \begin{bmatrix} 1 & 1 & 1 & \cdots \\ 1 & \omega^{-1} & \omega^{-2} & \cdots \\ 1 & \omega^{-2} & \omega^{-4} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Is invertible with an inverse:

$$M^{-1} = \frac{1}{n}\begin{bmatrix} 1 & 1 & 1 & \cdots \\ 1 & \omega & \omega^2 & \cdots \\ 1 & \omega^2 & \omega^4 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Definition 1.1.1. (Discrete Fourier Transform) The linear map $\varphi : \mathbb{C}^n \to \mathbb{C}^n$

given by the matrix

$$M = \frac{1}{n}\begin{bmatrix} 1 & 1 & 1 & \cdots \\ 1 & \omega^{-1} & \omega^{-2} & \cdots \\ 1 & \omega^{-2} & \omega^{-4} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

is called the discrete fourier transform.

1.1.2 Fourier Series

Definition 1.1.2. ($\mathbb{S}$) Denote $\mathbb{S}$ as the space of all complex valued $\mathbb{Z}$ periodic functions on $\mathbb{R}$ that are Lebesgue Measurable (For the applications, we take them to be *continuous*). If $f \in \mathbb{S}$, then $\forall\ x \in \mathbb{R},\ f(x) = f(x+1)$

Definition 1.1.3. (Hermitian Inner Product) On $\mathbb{S}$ (Which is a vector space), define $\langle f, g\rangle := \int_0^1 f\bar{g}dx$

Definition 1.1.4. (Fourier Coefficients c_n and the Fourier Series) For $n \in \mathbb{Z}$

$$c_n(f) := \langle f, \exp(2\pi inx)\rangle = \int_0^1 f \exp(-2\pi inx)dx$$

The Fourier Series of the function f is defined as

$$\sum_{-\infty}^{\infty} c_n \exp(2\pi inx) = \sum_{n=-\infty}^{\infty} (\int_0^1 f(t)\exp(-2\pi int)dt)(\exp(2\pi inx))$$

Lemma 1.1.5. *Let $f : \mathbb{R} \to \mathbb{C}$ be a $\mathbb{Z}$ periodic function. Let $c_n := \langle f, \exp(2\pi inx)\rangle$ as usual. Call f "nice" if $\{c_n\}$ converges abosolutely, $|c_n| \leq C/n^2$ for some constant C and for some $e > 1/2$, for all $x \in \mathbb{R}, h \in \mathbb{R}^+$,*

$$|f(x+h) - f(x)| \leq C|h|^e$$

(e=1 is called Lipshitz*).*
If f is "nice" and $c_n(f) = \langle f(x), \exp(2\pi inx)\rangle = 0$ for all $n \in \mathbb{N}$, then $f \equiv 0$.

Lemma 1.1.6. *If $f : \mathbb{R} \to \mathbb{C}$ (a $\mathbb{Z}$ periodic function) is "nice" (as defined in 1.1.5), then*

$$\sum_{n\in\mathbb{Z}} c_n \exp(2\pi inx) = f(x)$$

Lemma 1.1.7. ***Parseval*** *Suppose $f : \mathbb{R} \to \mathbb{C}$ is a "nice" function (as defined in 1.1.5). Then*

$$\langle f, f\rangle = \sum_{n\in\mathbb{Z}} |c_n|^2$$

Proof. $\langle f, f\rangle = \int_0^1 (\sum_{n\in\mathbb{Z}} c_n \exp(2\pi i n x))(\sum_{m\in\mathbb{Z}} c_m \exp(-2\pi m x))dx$ which gives

$$\sum_n \sum_n c_n c_m \int_0^1 (\exp(2\pi i x(n-m))dx)$$

which yields

$$\sum_n \sum_m c_n c_m \delta_{mn} = \sum_n |c_n|^2$$

□

A simple example with $\zeta-$function

We define

$$\zeta(s) := \sum_{n=1}^{\infty} 1/n^s \tag{1.1.3}$$

This converges for all $s > 1$ and diverges for all $s \leq 1$ Let $f(x) = 2\pi^2(x - 1/2)^2$. By computation we have $\langle f, f\rangle = \pi^4/20$, $c_0 = 2\pi^2/6$ and $c_n = 1/n^2$. Parseval gives us

$$\sum_{n\in\mathbb{Z}} |c_n|^2 = \langle f, f\rangle = \pi^2/20$$

Which in turn gives us that

$$\zeta(4) = \pi^4/90$$

Cauchy's Theorem

Theorem 1.1.8. *(**Cauchy- weak version**) Let $f : U \to \mathbb{C}$ be holomorphic on the simply connected domain U. Let P be some polygon on U. Then*

$$\int_P f(z)dz = 0$$

1.2 Modular Forms

Definition 1.2.1. (Upper Half Plane $\mathbb{H}$)

$$\mathbb{H} := \{z \in \mathbb{C} : im(z) > 0\}$$

Definition 1.2.2. ($SL(2,\mathbb{R})$)

$$SL(2,\mathbb{R}) := \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \textbf{ and } ad - bc = 1 \right\}$$

Definition 1.2.3. ($GL(2,\mathbb{R})$)

$$GL(2,\mathbb{R}) := \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \textbf{ and } ad - bc \neq 0 \right\}$$

Definition 1.2.4. ($GL^+(2,\mathbb{R})$)

$$GL^+(2,\mathbb{R}) := \left\{ M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a,b,c,d \in \mathbb{R} \textbf{ and } ad - bc > 0 \right\}$$

We have rather trivially the following inclusions

$$SL(2,\mathbb{R}) \hookrightarrow GL^+(2,\mathbb{R}) \hookrightarrow GL(2,\mathbb{R})$$

Definition 1.2.5. (Modular Form) Let $k \in \mathbb{Z}$. $f : \mathbb{H} \to \mathbb{C}$ is a modular form of weight k if:

1. For all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2,\mathbb{Z})$ and $z \in \mathbb{H}$,

$$f(\frac{az+b}{cz+d}) = (cz+d)^k f(z)$$

2. $f(z)$ is bounded as $im(z) \to \infty$
3. f is Holomorphic on all of $\mathbb{H}$

Corollary 1.2.6. $\forall x \in \mathbb{H}\ f(x+1) = f(x)$

Proof. Since $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in SL(2,\mathbb{Z})$, we have the result trivially.

□

Corollary 1.2.7. $\forall x \in \mathbb{H}\ f(-1/x) = x^k f(x)$

Proof. Since $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in SL(2, \mathbb{Z})$, we have the result.

□

Corollary 1.2.8. $f(x) = (-1)^k f(x)$

Proof. Since $\begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \in SL(2, \mathbb{Z})$, the result holds. Moreover, if k is odd, f has to be the 0 map. ***The only non zero modular forms are of even weights.***

□

Fact 1.2.9. $SL(2, \mathbb{Z})$ is generated by $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$.

Corollary 1.2.10. *From 1.2.9 we have that any modular form can be re-characterised (without reference to $SL(2, \mathbb{Z})$) by the following axioms: $f : \mathbb{H} \to \mathbb{C}$ is a Modular Form of weight k if:*

1. *f is holomorphic on $\mathbb{H}$*
2. *$f(\tau)$ bounded as $im(\tau) \to \infty$ (bounded along the y coordinate of the complex plane)*
3. *$f(\tau + 1) = f(\tau)$ and $f(-1/\tau) = (\tau)^k f(\tau)$ $\forall \tau \in \mathbb{H}$*

Corollary 1.2.11. *Since f is $\mathbb{Z}$ periodic, and holomorphic, we can fourier expand f, which gives*

$$f(x + iy) = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n(x+iy)} = \sum_{n \in \mathbb{Z}} a_n e^{2\pi i n x} e^{-2\pi n y}$$

*If a_n for negative n were to be non zero, we would have that f is unbounded as $y \to \infty$ (since $e^{2\pi n y}$ would go unbounded). Therefore, $\forall n < 0, a_n = 0$. a_0 can be 0, in which case $\lim_{y \to \infty} f(x + iy) = a_0 = 0$. If $a_0 = 0$, then the modular form is called a **Cusp Form**.*

Weakly Holomorphic Modular Forms of weight k

Definition 1.2.12. **(Weakly Holomorphic Modular Forms of weight k)** We say a function $f : \mathbb{H} \to \mathbb{C}$ is a weakly holomorphic modular form of weight k if:

1. $f(a\tau + b/c\tau + d) = (c\tau + d)^k f(\tau)$ for every $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z})$

2. $f(\tau) \in \mathcal{O}(q^{-N})$ instead of $\mathcal{O}(q^0)$ (for $f = \sum_{n \in \mathbb{Z}} a_n q^n$) which yields $f = \sum_{n=-N}^{\infty} a_n q^n$ (Note that f loses the boundedness property as $im(x)$ goes to infinity.)

Appendices